



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
FALL SEMESTER 2026-2027

Programme Name & Branch : M. Tech CSE and All Specialization
Course Code and Course Name : MACSE518 - Database Modeling and Design
Faculty Name(s) : Prof. Saravanaguru RA.K and Prof. Ramanathan L
Class Number(s) : VL2026270106053 and VL2026270106063
Date of Examination : 12/08/2026 & 2 PM - 3.30 PM
Exam Duration : 90 minutes **Maximum Marks: 50**

General instruction(s):

- Answer All Questions
- Course Outcomes:
CO1 - Comprehend the fundamental of database systems and its architectures.
CO2 - Model the database systems using ER and EER to design complex database systems.

Q. No	Question	M	CO	BL
1.	As businesses continue to transition away from conventional file-processing methods toward database solutions for data handling, critically analyze the fundamental characteristics that set the database approach apart from its file-based predecessors.	10	1	2
2.	Design an Extended Entity-Relationship (EER) diagram for a university system to manage information about academic departments, professors, deans, and department chairs. The database should capture: • Department details including name, campus location, and annual budget. • Professor information such as name, academic rank (e.g., Assistant, Associate, Full), and primary research area. • Dean details like name and years of administrative service. • Chair details like name and term duration (in years) Draw the complete EER model by addressing the following sub-questions in order: (a) Create the parent (superclass) group (b) Define the two main child groups (c) Break down the Administrator group further (d) Draw the full ISA (parent-child) hierarchy (e) Explain the constraints (rules) at each level	10	2	2
3	Imagine you are architecting a blockchain-driven banking database for a FinTech startup. The system serves three personas—customers checking real-time balances/transactions via an app, auditors reviewing immutable historical logs, and developers optimizing physical storage. Leveraging the ANSI/SPARC three-level schema framework (external, conceptual, internal), demonstrate a design that: prevents customers from accessing audit trails or storage metadata; isolates historical querying so it never stalls live transaction processing; and abstracts physical storage changes (e.g., transitioning from SSD to NVMe) so they are transparent to all front-end queries. Finally, connect each schema layer to logical	10	1	2



VIT

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	and physical data independence, and explain how this shields the system from future hardware or application evolution.			
4.	Aircraft (<u>aid</u>, aname, total_capacity, range) Certified (<u>cid</u>, <u>aid</u>) Employees (<u>eid</u>, ename, salary) Consider the above schema and write the following query into relational algebra. The key attribute is underlined. (i) List the names of pilots certified for some Boeing aircraft. (ii) List the flight number that can be piloted by every pilot whose salary is more than Rs.1,50,000. (iii) Find the total amount paid as salaries to Boeing certified pilot. (iv) Find the names of pilots who can operate planes with a range greater than 3,000 miles but are not certified on any Boeing aircraft. (v) Find the aids of all aircraft that can be used on non-stop flights from Chennai to Delhi. (A flight will be considered as non-stop, if the aircraft range is greater than equal to the distance of flight).	10	2	3
5.	Product (prod_id, prod_name, mfg_date, category, supp_id) Supplier (supp_id, supp_name, city) Write the SQL command for the below queries. 1.Show products that are not supplied by any supplier. 2.Display products manufactured in or after 2023. 3.List the product IDs of 'Electronics' and 'Furniture' items from supplier id 201. 4.Count the number of products supplied by each supplier (show supplier name and count). 5.List all 'Food' items manufactured before 2022.	10	2	3
